

The AI Landscape

Model Selection, Open vs Closed Models, Decision Matrix & Modern AI Tech Stack

Instructor & Student Guide · Engineering Tradeoffs · Production Architecture

*Part 3 Overview: Picking an AI model is an **engineering trade-off decision**, not a benchmark leaderboard contest. This module details the critical differences between open-weight and closed-API models, provides a 6-dimensional model selection matrix, maps where AI lives in your software stack, and recaps the complete lifecycle of artificial intelligence.*

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10. The AI Landscape: Model Selection & Developer Stack

1.1 Open Models vs Closed API Models

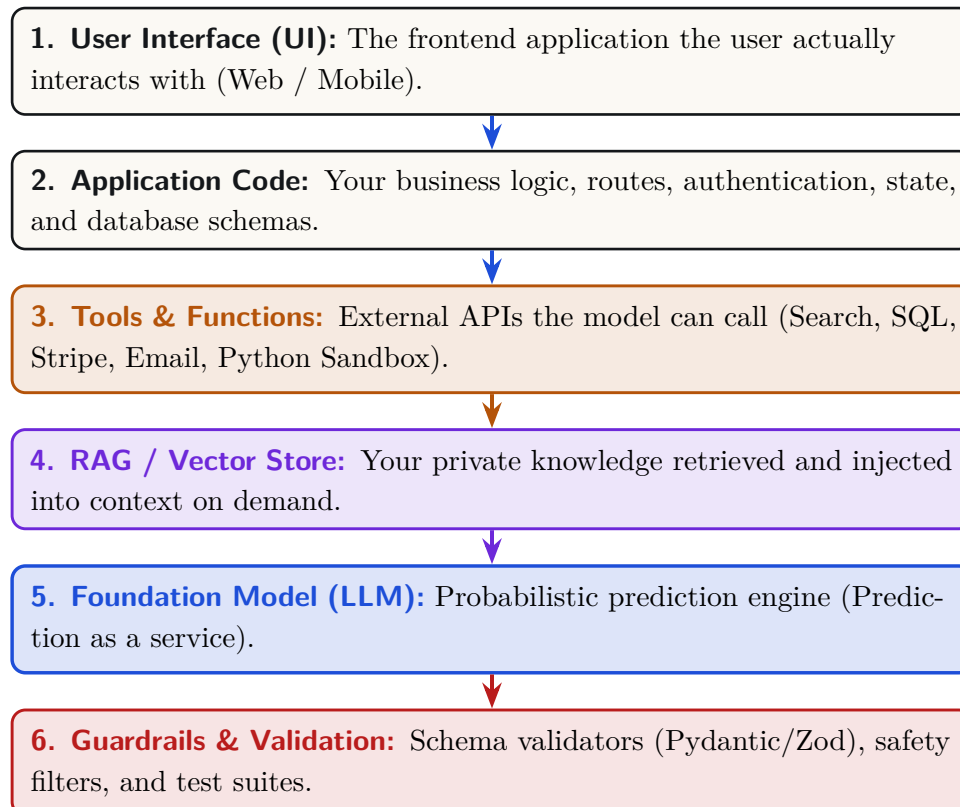
Dimension	Open Models (Llama, Mistral, Gemma)	Closed API Models (GPT-4o, Claude 3.5, Gemini)
Ownership	You download and hold the weight files.	You query a managed HTTP API endpoint.
Data Privacy	100% private: data never leaves your VPC/hardware.	Data traverses vendor networks (governed by API terms).
Customization	Full access: fine-tune, modify layers, quantize.	Constrained to vendor-supported fine-tuning APIs.
Operations	You own GPU provisioning, uptime, and scaling.	Zero infrastructure maintenance.
Pricing Model	Fixed compute cost (GPU rental/purchase).	Variable cost (billed per token indefinitely).
Failure Mode	You manage your own server crashes and 3 AM alerts.	Subject to third-party rate limits, outages, and policy shifts.

1.2 The 6-Dimensional Model Decision Matrix

Dimension	Small Model (7B)	Mid-Size Model (70B)	Frontier Model (400B+)
Cost	Lowest: cheap enough for millions of daily queries.	Moderate: balanced cost-per-call for standard apps.	Highest: expensive; reserved for complex high-value tasks.
Latency	Fastest (sub-100ms): ideal for autocomplete/UX.	Balanced: suitable for interactive streaming chat.	Slowest: seconds per response; best for async jobs.
Capability	Narrow: strong on repetitive tasks after tuning.	Broad: handles 80% of enterprise drafting and extraction.	Broadest: deep logic, complex coding, and multi-step reasoning.
Deployment	Laptop, single GPU, or on-device edge hardware.	Single multi-GPU server (e.g., 4× or 8× A100).	Managed API or massive sharded multi-node GPU cluster.
Privacy	Full control: run completely offline or on-premise.	Self-hosted inside your private cloud VPC.	Vendor API terms and regional compliance boundaries.
Throughput	High volume: millions of requests at minimal cost.	Steady production application traffic.	Selective routing: small model first, escalate if needed.

1.3 Where AI Sits in the Modern Software Stack

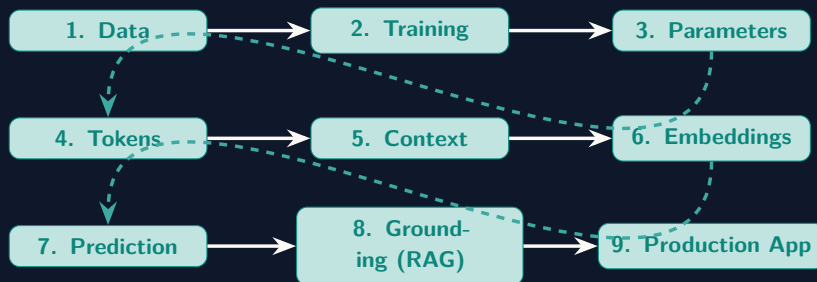
AI is not replacing your software stack. It becomes another layer inside it. You still write code, handle auth, manage databases, write tests, and deploy APIs. The model is simply a probabilistic component you call with unusual properties.



2 The Grand Recap: The Complete AI Lifecycle

The Complete 9-Stage AI Lifecycle Pipeline

QUICK REVISION & MASTER TAKEAWAYS



2.1 Master Glossary of Key Terms

Term	Category	One-Line Technical Definition
Parameter	Model Architecture	A learned floating-point number (weight) adjusted during training.
Token	Data Processing	Subword chunk (≈ 4 characters) converted to an integer ID.
Context Window	Inference	The temporary active token capacity processed in a single interaction.
Embedding	Vector Representation	High-dimensional coordinate array encoding semantic meaning.
Cosine Similarity	Vector Search	Mathematical metric measuring the semantic angle between two embedding vectors.
RAG	System Architecture	Fetching verified external documents and injecting them into context at query time.
Hallucination	Model Behavior	Statistically plausible yet factually incorrect output generated by probabilistic sampling.
Quantization	Optimization	Reducing weight precision ($\text{FP16} \rightarrow \text{INT8}/\text{INT4}$) to cut VRAM and speed up inference.
Fine-Tuning	Adaptation	Updating pre-trained weights for specific formats, styles, or tasks.
Open Model	Deployment	Model whose weights are publicly downloadable and self-hostable.

- **THE FINAL 10-SECOND MENTAL MODEL TO REMEMBER FOREVER**

AI applications are **standard software systems** wrapping a probabilistic prediction component. The model provides intelligence; your code provides structure, data, retrieval, tools, and verification.